

MINI LAB: GROUP ASSIGNMENT

Group Name:

- Almira Shinta Aulia
- Melinia Fransiska Telaumbanua
- Naura Rizky
- Shofiyatur Rahmatin Nazilah

Lab 1: Network Reconnaissance (PPT Week 4)

1. Scan using nmap find the host (assume you don't know the linux ip address) from your linux friend.

- What is the command ? Why ?

```
memeelup@memeelup-VirtualBox:~$ nmap -Pn -PR 192.168.43.0
Starting Nmap 7.94SVN ( https://nmap.org ) at 2025-11-18 15:47 WIB
Nmap scan report for 192.168.43.0
Host is up (0.80s latency).
All 1000 scanned ports on 192.168.43.0 are in ignored states.
Not shown: 980 filtered tcp ports (no-response), 20 filtered tcp ports (host-unr
each)

Nmap done: 1 IP address (1 host up) scanned in 34.96 seconds
```

2. Scan using nmap with some options

```
memeelup@memeelup-VirtualBox:~$ sudo su
root@memeelup-VirtualBox:/home/memeelup# nmap -A 192.168.43.0
Starting Nmap 7.94SVN ( https://nmap.org ) at 2025-11-18 16:06 WIB
Note: Host seems down. If it is really up, but blocking our ping probes, try -Pn
Nmap done: 1 IP address (0 hosts up) scanned in 2.46 seconds
root@memeelup-VirtualBox:/home/memeelup# exit
exit
memeelup@memeelup-VirtualBox:~$ nmap -A 192.168.43.0
Starting Nmap 7.94SVN ( https://nmap.org ) at 2025-11-18 16:07 WIB
Note: Host seems down. If it is really up, but blocking our ping probes, try -Pn
Nmap done: 1 IP address (0 hosts up) scanned in 3.36 seconds
memeelup@memeelup-VirtualBox:~$ nmap -sV 192.168.43.0
Starting Nmap 7.94SVN ( https://nmap.org ) at 2025-11-18 16:08 WIB
Note: Host seems down. If it is really up, but blocking our ping probes, try -Pn
Nmap done: 1 IP address (0 hosts up) scanned in 3.28 seconds
memeelup@memeelup-VirtualBox:~$ nmap -Pn -sV 192.168.43.0
Starting Nmap 7.94SVN ( https://nmap.org ) at 2025-11-18 16:09 WIB

memeelup@memeelup-VirtualBox:~$ nmap -Pn -sV 192.168.43.0
Starting Nmap 7.94SVN ( https://nmap.org ) at 2025-11-18 16:09 WIB
Nmap scan report for 192.168.43.0
Host is up (0.86s latency).
All 1000 scanned ports on 192.168.43.0 are in ignored states.
Not shown: 980 filtered tcp ports (no-response), 20 filtered tcp ports (host-unreach)

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 44.87 seconds
```

3. What is your linux existing services ?

```
memeelup@memeelup-VirtualBox:~$ nmap -sU 192.168.43.0
You requested a scan type which requires root privileges.
QUITTING!
memeelup@memeelup-VirtualBox:~$ sudo nmap -sU 192.168.43.0
[sudo] password for memeelup:
Starting Nmap 7.94SVN ( https://nmap.org ) at 2025-11-18 15:55 WIB
Note: Host seems down. If it is really up, but blocking our ping probes, try -Pn
Nmap done: 1 IP address (0 hosts up) scanned in 1.88 seconds
memeelup@memeelup-VirtualBox:~$ sudo nmap -Pn -sU 192.168.43.0
Starting Nmap 7.94SVN ( https://nmap.org ) at 2025-11-18 15:55 WIB
Nmap done: 1 IP address (0 hosts up) scanned in 1.59 seconds
memeelup@memeelup-VirtualBox:~$ sudo nmap -Pn -F 192.168.43.0
Starting Nmap 7.94SVN ( https://nmap.org ) at 2025-11-18 15:57 WIB
Nmap done: 1 IP address (0 hosts up) scanned in 2.01 seconds
memeelup@memeelup-VirtualBox:~$ nmap -Pn -F 192.168.43.0
Starting Nmap 7.94SVN ( https://nmap.org ) at 2025-11-18 15:58 WIB
Nmap scan report for 192.168.43.0
Host is up (0.074s latency).
All 100 scanned ports on 192.168.43.0 are in ignored states.
Not shown: 90 filtered tcp ports (no-response), 10 filtered tcp ports (host-unreach)

Nmap done: 1 IP address (1 host up) scanned in 3.54 seconds
memeelup@memeelup-VirtualBox:~$ nmap -sL 192.168.43.0
Starting Nmap 7.94SVN ( https://nmap.org ) at 2025-11-18 16:02 WIB
Nmap scan report for 192.168.43.0
Nmap done: 1 IP address (0 hosts up) scanned in 0.13 seconds
memeelup@memeelup-VirtualBox:~$ nmap -Pn -sL 192.168.43.0
Starting Nmap 7.94SVN ( https://nmap.org ) at 2025-11-18 16:03 WIB
Nmap scan report for 192.168.43.0
Nmap done: 1 IP address (0 hosts up) scanned in 3.62 seconds
```

4. Create http and ftp service below

```
memeelup@memeelup-VirtualBox:~$ sudo nmap -Pn -sS 192.168.43.0/24
Starting Nmap 7.94SVN ( https://nmap.org ) at 2025-11-25 15:02 WIB
Nmap scan report for LAPTOP-N1JQ6BC9 (192.168.43.86)
Host is up (0.0041s latency).
All 1000 scanned ports on LAPTOP-N1JQ6BC9 (192.168.43.86) are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)
MAC Address: F4:6A:DD:6A:B7:8B (Liteon Technology)

Nmap scan report for memeelup-VirtualBox (192.168.43.9)
Host is up (0.0000080s latency).
Not shown: 997 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
80/tcp    open  http
```

5. What is the port for http ?

HTTP Port: 80/tcp

```
memeelup@memeelup-VirtualBox:~$ sudo nmap -Pn -sS 192.168.43.0/24
Starting Nmap 7.94SVN ( https://nmap.org ) at 2025-11-25 15:02 WIB
Nmap scan report for LAPTOP-N1JQ6BC9 (192.168.43.86)
Host is up (0.0041s latency).
All 1000 scanned ports on LAPTOP-N1JQ6BC9 (192.168.43.86) are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)
MAC Address: F4:6A:DD:6A:B7:8B (Liteon Technology)

Nmap scan report for memeelup-VirtualBox (192.168.43.9)
Host is up (0.0000080s latency).
Not shown: 997 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
80/tcp    open  http
```

6. What is/are the port/s for ftp?

- Screenshot from your linux using command netstat -tulpn

- Screenshot from nmap

FTP Port: 21/tcp

```
memeelup@memeelup-VirtualBox:~$ sudo nmap -Pn -sS 192.168.43.0/24
Starting Nmap 7.94SVN ( https://nmap.org ) at 2025-11-25 15:02 WIB
Nmap scan report for LAPTOP-N1JQ6BC9 (192.168.43.86)
Host is up (0.0041s latency).
All 1000 scanned ports on LAPTOP-N1JQ6BC9 (192.168.43.86) are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)
MAC Address: F4:6A:DD:6A:B7:8B (Liteon Technology)

Nmap scan report for memeelup-VirtualBox (192.168.43.9)
Host is up (0.0000080s latency).
Not shown: 997 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
80/tcp    open  http
```



```
memeelup@memeelup-VirtualBox:~$ netstat -tulpn
(Not all processes could be identified, non-owned process info
 will not be shown, you would have to be root to see it all.)
Active Internet connections (only servers)
Proto Recv-Q Send-Q Local Address           Foreign Address         State       PID/Program name
tcp        0      0 0.0.0.0:22             0.0.0.0:*               LISTEN      -
tcp        0      0 127.0.0.54:53          0.0.0.0:*               LISTEN      -
tcp        0      0 127.0.0.53:53          0.0.0.0:*               LISTEN      -
tcp        0      0 127.0.0.1:631          0.0.0.0:*               LISTEN      -
tcp6       0      0 :::80                  :::*                     LISTEN      -
tcp6       0      0 :::21                  :::*                     LISTEN      -
tcp6       0      0 :::22                  :::*                     LISTEN      -
tcp6       0      0 :::1631                 :::*                     LISTEN      -
udp        0      0 127.0.0.54:53          0.0.0.0:*               -           -
udp        0      0 127.0.0.53:53          0.0.0.0:*               -           -
udp        0      0 0.0.0.0:50403          0.0.0.0:*               -           -
udp        0      0 0.0.0.0:5353           0.0.0.0:*               -           -
udp6       0      0 :::42885                :::*                     -           -
udp6       0      0 :::5353                 :::*                     -           -
```

7. Stop ftp services.

- Screenshot from nmap

```
memeelup@memeelup-VirtualBox:~$ service vsftpd stop
memeelup@memeelup-VirtualBox:~$ sudo nmap -Pn -sS 192.168.43.0/24
Starting Nmap 7.94SVN ( https://nmap.org ) at 2025-11-25 15:11 WIB
Nmap scan report for LAPTOP-N1JQ6BC9 (192.168.43.86)
Host is up (0.00067s latency).
All 1000 scanned ports on LAPTOP-N1JQ6BC9 (192.168.43.86) are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)
MAC Address: F4:6A:DD:6A:B7:8B (Liteon Technology)

Nmap scan report for memeelup-VirtualBox (192.168.43.9)
Host is up (0.000011s latency).
Not shown: 998 closed tcp ports (reset)
PORT      STATE SERVICE
22/tcp    open  ssh
80/tcp    open  http
```

8. Screenshot your windows access your linux web

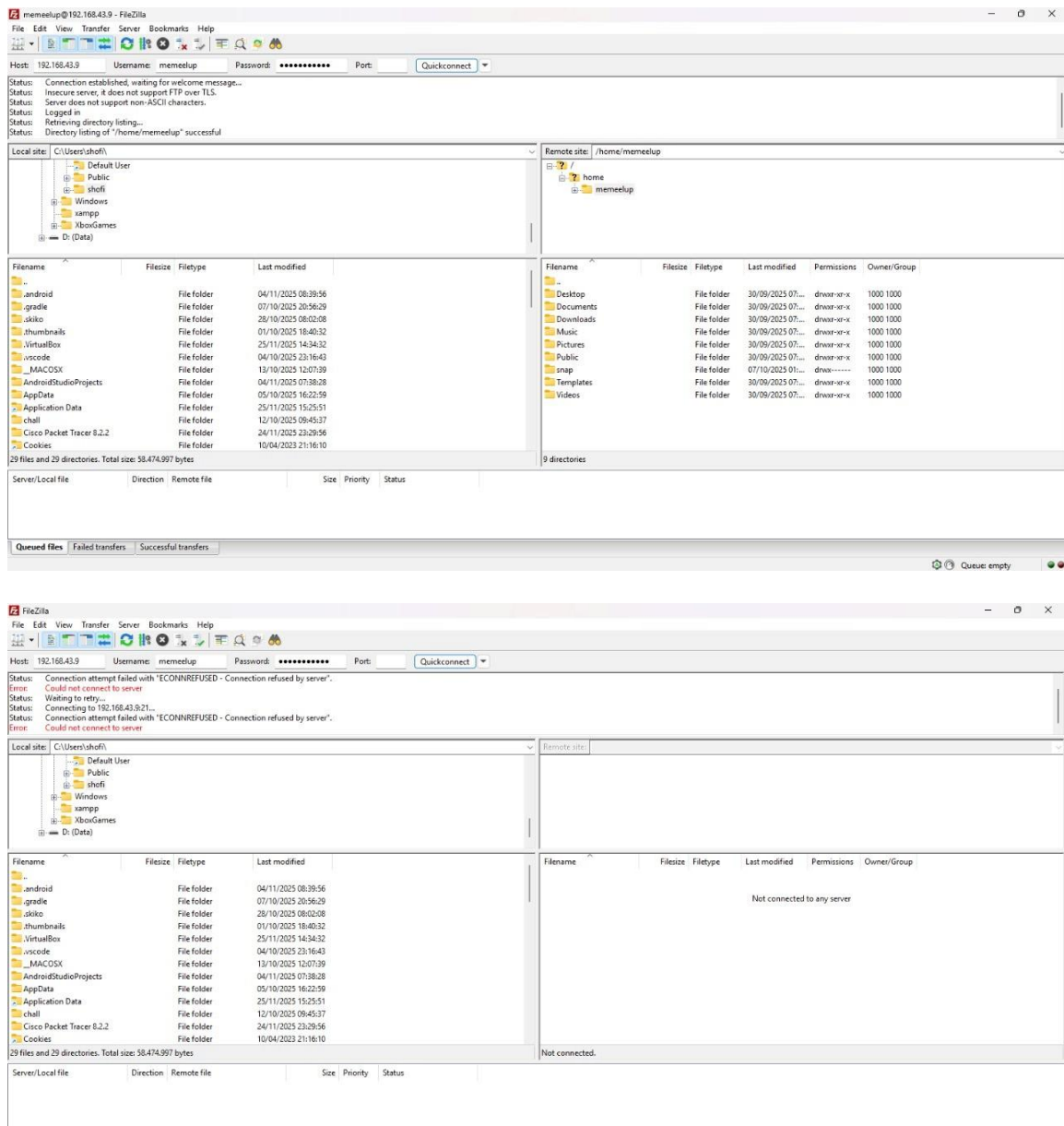
The screenshot shows a web browser window with the address bar displaying '192.168.43.9'. The page content includes the Ubuntu logo, the title 'Apache2 Default Page', and a message 'It works!'. Below this, there is a paragraph of text explaining the purpose of the page and a section titled 'Configuration Overview' which contains a code block showing the contents of the /etc/apache2/apache2.conf file.

```

/etc/apache2/
|-- apache2.conf
|   |-- ports.conf
|   |-- mods-enabled
|   |   |-- *.load
|   |   |-- *.conf
|   |-- conf-enabled
|   |   |-- *.conf
|   |-- sites-enabled
|   |   |-- *.conf

```

9. Screenshoot your filezilla access your linux ftp

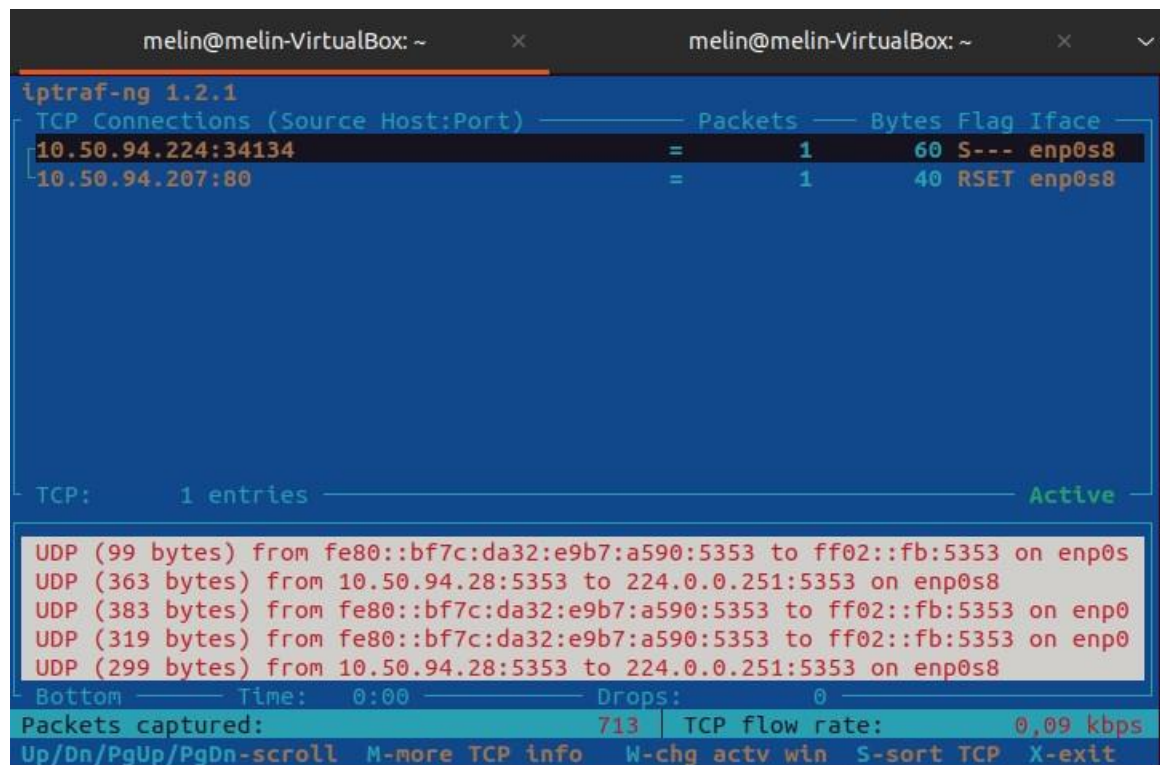


10. If we see iptraf like this, it warns you that somebody sniff your computer. Why ?

When iptraf displays many connections originating from the same IP address to many different ports very quickly, and all of those connections end with RST, it means that someone is scanning your computer. Normal traffic never behaves like that. Attackers use port scanning tools that quickly “knock” on many ports to see which ones are open. This scanning is usually done before monitoring or attempting to attack your system.

Therefore, iptraf displays a warning because it shows clear signs that someone is testing your computer.

11. Screenshoot your iptraf at linux when your server is scanned by nmap.



The screenshot shows the iptraf-ng 1.2.1 interface. At the top, there are two tabs labeled 'melin@melin-VirtualBox: ~'. The main window displays a table of TCP connections:

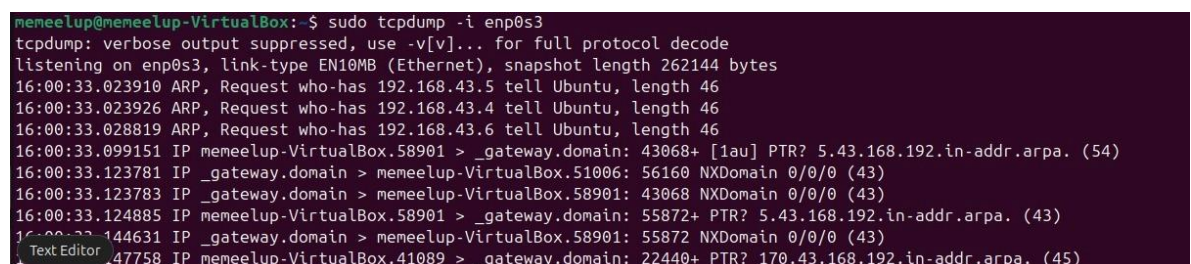
TCP Connections (Source Host:Port)	Packets	Bytes	Flag	Iface
10.50.94.224:34134	= 1	60	S---	enp0s8
10.50.94.207:80	= 1	40	RSET	enp0s8

Below the table, it shows 'TCP: 1 entries' and 'Active'. A scrollable area displays UDP traffic:

- UDP (99 bytes) from fe80::bf7c:da32:e9b7:a590:5353 to ff02::fb:5353 on enp0s
- UDP (363 bytes) from 10.50.94.28:5353 to 224.0.0.251:5353 on enp0s8
- UDP (383 bytes) from fe80::bf7c:da32:e9b7:a590:5353 to ff02::fb:5353 on enp0
- UDP (319 bytes) from fe80::bf7c:da32:e9b7:a590:5353 to ff02::fb:5353 on enp0
- UDP (299 bytes) from 10.50.94.28:5353 to 224.0.0.251:5353 on enp0s8

At the bottom, a status bar shows 'Packets captured: 713' and 'TCP flow rate: 0,09 kbps'. Navigation keys are listed: 'Up/Dn/PgUp/PgDn-scroll', 'M-more TCP info', 'W-chg actv win', 'S-sort TCP', and 'X-exit'.

12. Screenshoot your tcpdump at linux when your server is scanned by nmap.



The screenshot shows the output of the command 'sudo tcpdump -i enp0s3' in a terminal window. The output is as follows:

```
memeelup@memeelup-VirtualBox:~$ sudo tcpdump -i enp0s3
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on enp0s3, link-type EN10MB (Ethernet), snapshot length 262144 bytes
16:00:33.023910 ARP, Request who-has 192.168.43.5 tell Ubuntu, length 46
16:00:33.023926 ARP, Request who-has 192.168.43.4 tell Ubuntu, length 46
16:00:33.028819 ARP, Request who-has 192.168.43.6 tell Ubuntu, length 46
16:00:33.099151 IP memeelup-VirtualBox.58901 > _gateway.domain: 43068+ [1au] PTR? 5.43.168.192.in-addr.arpa. (54)
16:00:33.123781 IP _gateway.domain > memeelup-VirtualBox.51006: 56160 NXDomain 0/0/0 (43)
16:00:33.123783 IP _gateway.domain > memeelup-VirtualBox.58901: 43068 NXDomain 0/0/0 (43)
16:00:33.124885 IP memeelup-VirtualBox.58901 > _gateway.domain: 55872+ PTR? 5.43.168.192.in-addr.arpa. (43)
16:00:33.144631 IP _gateway.domain > memeelup-VirtualBox.58901: 55872 NXDomain 0/0/0 (43)
16:00:33.147758 IP memeelup-VirtualBox.41089 > _gateway.domain: 22440+ PTR? 170.43.168.192.in-addr.arpa. (45)
```

Lab 2: 2Attack Simulation (PPT Week 6)

-ip address attacker : 192.168.1.117

-ip address victim : 192.168.1.116

1. Create the above configuration (your linux friend connect to your linux)

Ip address attacker :

```
melin@melin-VirtualBox:~$ ip route show
default via 10.0.2.2 dev enp0s3 proto dhcp metric 100
default via 192.168.1.1 dev enp0s8 proto dhcp metric 101
10.0.2.0/24 dev enp0s3 proto kernel scope link src 10.0.2.15 metric 100
169.254.0.0/16 dev enp0s3 scope link metric 1000
192.168.1.0/24 dev enp0s8 proto kernel scope link src 192.168.1.117 metric 101
```

Ip address victim :

```
winwin@Ubuntu:~$ ip route show
default via 10.0.2.2 dev enp0s3 proto dhcp src 10.0.2.15 metric 100
default via 192.168.1.1 dev enp0s9 proto dhcp src 192.168.1.116 metric 101
10.0.2.0/24 dev enp0s3 proto kernel scope link src 10.0.2.15 metric 100
192.168.1.0/24 dev enp0s9 proto kernel scope link src 192.168.1.116 metric 101
```

2. Screenshot above result (before attack) nload normal

```
Device enp0s9 [192.168.1.116] (1/1):
=====
Incoming:

Curr: 0.00 Bit/s
Avg: 56.00 Bit/s
Min: 0.00 Bit/s
Max: 472.00 Bit/s
Ttl: 46.75 kByte

Outgoing:

Curr: 0.00 Bit/s
Avg: 56.00 Bit/s
Min: 0.00 Bit/s
Max: 472.00 Bit/s
Ttl: 15.13 kByte
```

iptraf normal


```

iptraf-ng 1.2.1
Statistics for enp0s9

```

	Total Packets	Total Bytes	Incoming Packets	Incoming Bytes	Outgoing Packets	Outgoing Bytes
Total:	0	0	0	0	0	0
IPv4:	0	0	0	0	0	0
IPv6:	0	0	0	0	0	0
TCP:	0	0	0	0	0	0
UDP:	0	0	0	0	0	0
ICMP:	0	0	0	0	0	0
Other IP:	0	0	0	0	0	0
Non-IP:	0	0	0	0	0	0
Broadcast:	0	0	0	0	0	0

```

Total rates:      0.00 kbps      Broadcast rates:      0.00 kbps
                  0 pps                      0 pps

Incoming rates:   0.00 kbps
                  0 pps

Outgoing rates:   0.00 kbps

Time: 0:00      0 pps      Drops: 0
X-exit

```

tcpdump normal

```

memeelup@memeelup-VirtualBox:~$ sudo tcpdump -i enp0s9 icmp -w /tmp/icmp_baseline.pcap &
[1] 5200

```

3. What is your control / command to handle smurf attack ?

```

memeelup@memeelup-VirtualBox:~$ sudo iptables -A INPUT -p icmp -i enp0s9 -s 192.168.1.11/ j DROP

```

4. What's the hping3 command if you want to request ping 1000 bytes ?

Command

```

melin@melin-VirtualBox:~$ sudo hping3 -1 -d 1000--flood 192.168.1.116
HPING 192.168.1.116 (enp0s8 192.168.1.116): icmp mode set, 28 headers + 1000 data bytes
len=1028 ip=192.168.1.116 ttl=64 id=33492 icmp_seq=0 rtt=3.0 ms
len=1028 ip=192.168.1.116 ttl=64 id=33863 icmp_seq=1 rtt=3.9 ms
len=1028 ip=192.168.1.116 ttl=64 id=34378 icmp_seq=2 rtt=1.9 ms
len=1028 ip=192.168.1.116 ttl=64 id=34513 icmp_seq=3 rtt=3.5 ms
len=1028 ip=192.168.1.116 ttl=64 id=35341 icmp_seq=4 rtt=3.0 ms
len=1028 ip=192.168.1.116 ttl=64 id=35666 icmp_seq=5 rtt=4.1 ms
len=1028 ip=192.168.1.116 ttl=64 id=36019 icmp_seq=6 rtt=1.8 ms
len=1028 ip=192.168.1.116 ttl=64 id=36564 icmp_seq=7 rtt=2.6 ms
len=1028 ip=192.168.1.116 ttl=64 id=36788 icmp_seq=8 rtt=2.8 ms

```

result


```

iptraf-ng 1.2.1
- TCP Connections (Source Host:Port) ----- Packets ----- Bytes Flag Iface -----

```

```

- TCP: 0 entries ----- Active -----

```

```

ICMP echo rply (1028 bytes) from 192.168.1.116 to 192.168.1.117 on enp0s9
ICMP echo req (1028 bytes) from 192.168.1.117 to 192.168.1.116 on enp0s9
ICMP echo rply (1028 bytes) from 192.168.1.116 to 192.168.1.117 on enp0s9
ICMP echo req (1028 bytes) from 192.168.1.117 to 192.168.1.116 on enp0s9
ICMP echo rply (1028 bytes) from 192.168.1.116 to 192.168.1.117 on enp0s9

```

```

Bottom ----- Time: 0:00 ----- Drops: 0 -----

```

5. Comparing nload before and under attack

```
Device enp0s9 [192.168.1.116] (1/1):
=====
Incoming:

Curr: 0.00 Bit/s
Avg: 56.00 Bit/s
Min: 0.00 Bit/s
Max: 472.00 Bit/s
Ttl: 46.75 kByte

Outgoing:

Curr: 0.00 Bit/s
Avg: 56.00 Bit/s
Min: 0.00 Bit/s
Max: 472.00 Bit/s
Ttl: 15.13 kByte
```

After attack :

Device enp0s9 [192.168.1.116] (1/1):

=====

Incoming:

Curr: 936.00 Bit/s
Avg: 1.01 kBit/s
Min: 0.00 Bit/s
Max: 17.19 kBit/s
Ttl: 163.72 kByte

Outgoing:

Curr: 464.00 Bit/s
Avg: 568.00 Bit/s
Min: 0.00 Bit/s
Max: 5.04 kBit/s
Ttl: 36.83 kByte

Device enp0s9 [192.168.1.116] (1/1):

=====

Incoming:

Curr: 1.49 MBit/s
Avg: 1.39 MBit/s
Min: 438.45 kBit/s
Max: 1.80 MBit/s
Ttl: 20.59 MByte

..... ..
#####|....|||###

Outgoing:

Curr: 472.00 Bit/s
Avg: 16.00 Bit/s
Min: 0.00 Bit/s
Max: 1.23 kBit/s
Ttl: 42.07 kByte

6. Comparing at under attack condition with and without iptables

implementation A. Without iptables:

icmp(iptraf attacked)

```
iptraf-ng 1.2.1
Statistics for enp0s9
```

	Total Packets	Total Bytes	Incoming Packets	Incoming Bytes	Outgoing Packets	Outgoing Bytes
Total:	15	558	7	322	8	236
IPv4:	15	432	7	196	8	236
IPv6:	0	0	0	0	0	0
TCP:	0	0	0	0	0	0
UDP:	0	0	0	0	0	0
ICMP:	14	392	7	196	7	196
Other IP:	1	40	0	0	1	40
Non-IP:	0	0	0	0	0	0
Broadcast:	0	0	0	0	0	0
Total rates:		0.65 kbps 1 pps			Broadcast rates:	0.00 kbps 0 pps
Incoming rates:		0.36 kbps 0 pps				
Outgoing rates:		0.28 kbps			IP checksum errors:	0
Time: 0:00		0 pps			Drops:	0

B. With Iptables

Smurf attack :

```
melin@melin-VirtualBox:~$ sudo hping3 -1 --flood -a 192.168.1.116 192.168.1.255
HPING 192.168.1.255 (enp0s8 192.168.1.255): icmp mode set, 28 headers + 0 data bytes
hping in flood mode, no replies will be shown
```

jtcpdump smurf :

```
memeelup@memeelup-VirtualBox:~$ sudo tcpdump -i enp0s9 icmp -w /tmp/icmp_baseline.pcap &
[4] 5520
```

iptraf smurf :

iptraf-ng 1.2.1

Statistics for enp0s9

	Total Packets	Total Bytes	Incoming Packets	Incoming Bytes	Outgoing Packets	Outgoing Bytes
Total:	9835	452410	9835	452410	0	0
IPv4:	9835	275380	9835	275380	0	0
IPv6:	0	0	0	0	0	0
TCP:	0	0	0	0	0	0
UDP:	0	0	0	0	0	0
ICMP:	9835	275380	9835	275380	0	0
Other IP:	0	0	0	0	0	0
Non-IP:	0	0	0	0	0	0
Broadcast:	9835	452410	9835	452410	0	0

Total rates: 542.90 kbps
1475 pps

Broadcast rates: 542.90 kbps
1475 pps

Incoming rates: 542.90 kbps
1475 pps

IP checksum errors: 0

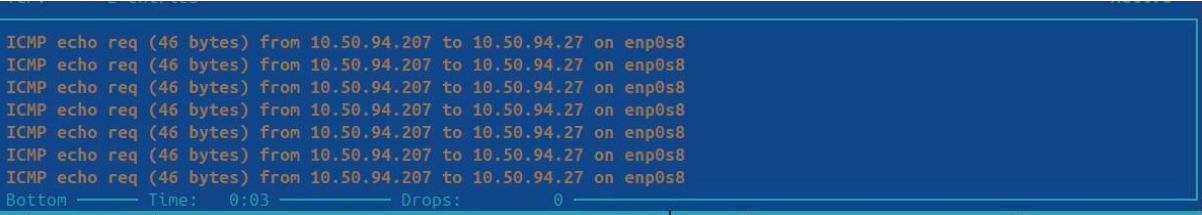
Outgoing rates: 0.00 kbps

Time: 0:00 0 pps Drops: 0

Lab 3: Firewall Configuration (PPT Week 7)

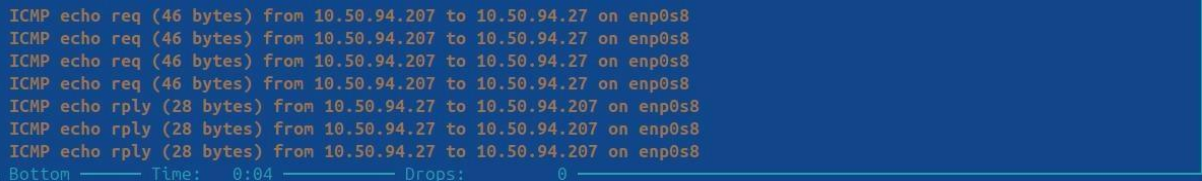
1. Complete this figure

```
fernandez@fernandez-VirtualBox:~$ sudo iptables -A INPUT -p icmp --icmp-type echo-request -j DROP
```



```
ICMP echo req (46 bytes) from 10.50.94.207 to 10.50.94.27 on enp0s8
ICMP echo req (46 bytes) from 10.50.94.207 to 10.50.94.27 on enp0s8
ICMP echo req (46 bytes) from 10.50.94.207 to 10.50.94.27 on enp0s8
ICMP echo req (46 bytes) from 10.50.94.207 to 10.50.94.27 on enp0s8
ICMP echo req (46 bytes) from 10.50.94.207 to 10.50.94.27 on enp0s8
ICMP echo req (46 bytes) from 10.50.94.207 to 10.50.94.27 on enp0s8
ICMP echo req (46 bytes) from 10.50.94.207 to 10.50.94.27 on enp0s8
Bottom Time: 0:03 Drops: 0
```

```
fernandez@fernandez-VirtualBox:~$ sudo iptables -F
```

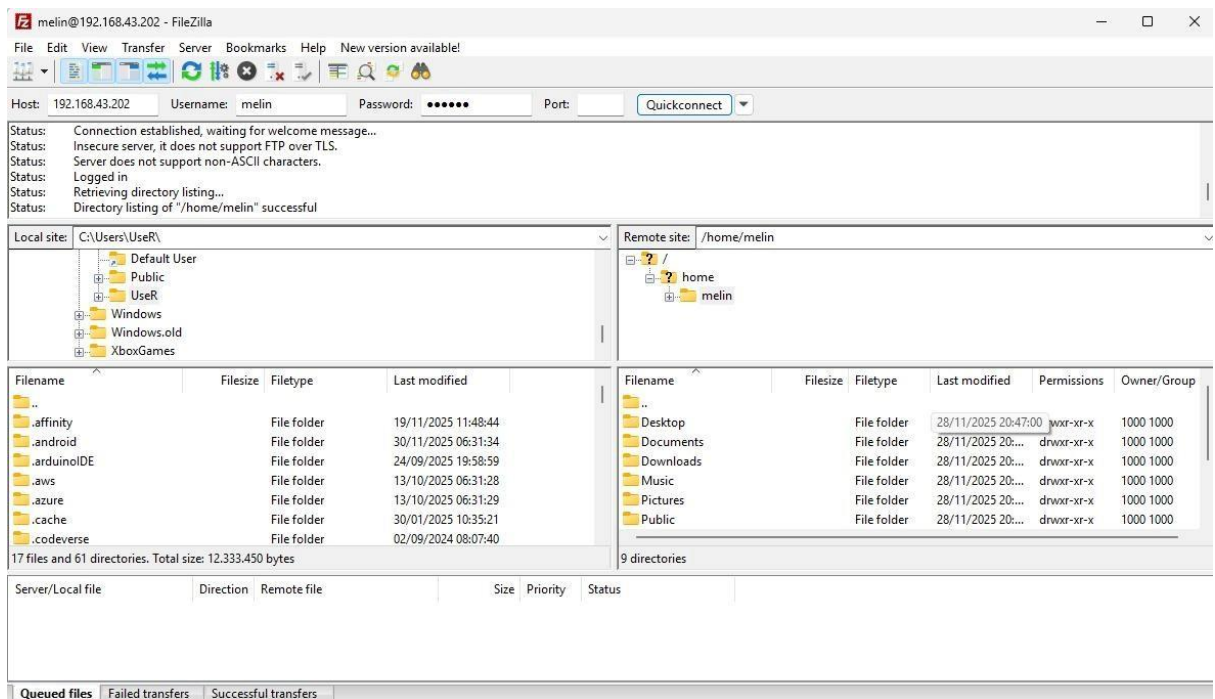


```
ICMP echo req (46 bytes) from 10.50.94.207 to 10.50.94.27 on enp0s8
ICMP echo req (46 bytes) from 10.50.94.207 to 10.50.94.27 on enp0s8
ICMP echo req (46 bytes) from 10.50.94.207 to 10.50.94.27 on enp0s8
ICMP echo req (46 bytes) from 10.50.94.207 to 10.50.94.27 on enp0s8
ICMP echo rply (28 bytes) from 10.50.94.27 to 10.50.94.207 on enp0s8
ICMP echo rply (28 bytes) from 10.50.94.27 to 10.50.94.207 on enp0s8
ICMP echo rply (28 bytes) from 10.50.94.27 to 10.50.94.207 on enp0s8
Bottom Time: 0:04 Drops: 0
```

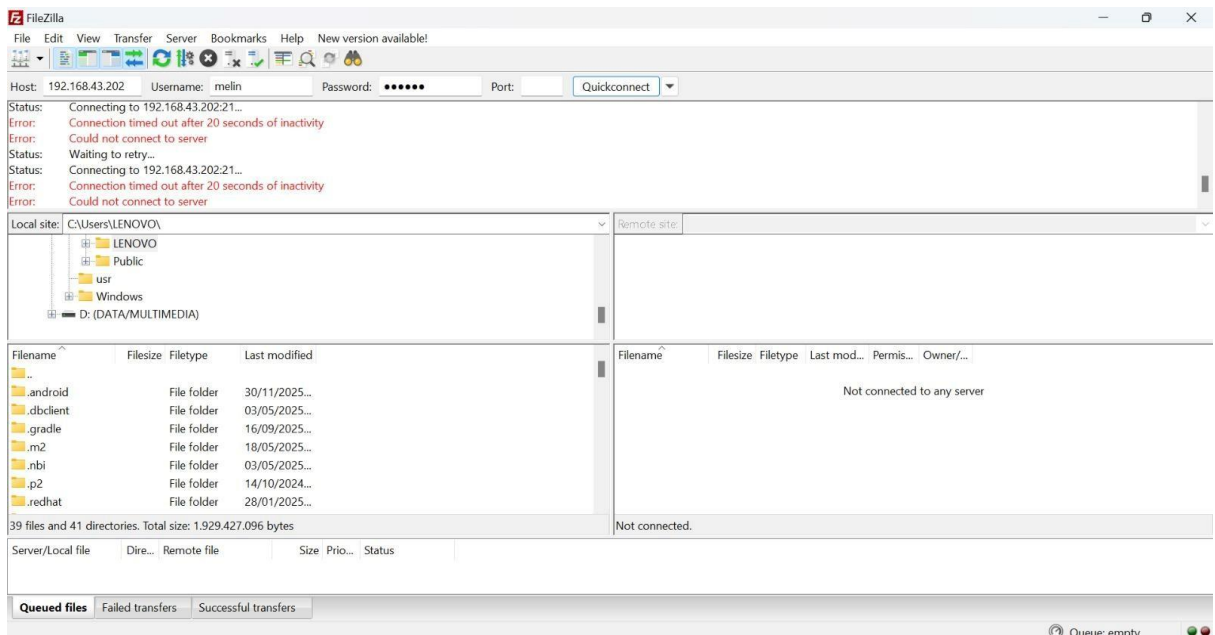
2. Your friends windows only possible to access web but not ftp. But Your windows possible to access web and ftp your friend windows:



your ftp connected (windows to linux):



your ftp connected but failed (windows to linux):



3. Your windows possible to get reply icmp but your friends windows cannot get reply icmp. Your friend linux possible to get reply icmp but cannot do ping attack to your linux. Windows

```
PS C:\Users\shofi> ping -t 192.168.43.211
```

```
Pinging 192.168.43.211 with 32 bytes of data:  
Reply from 192.168.43.86: Destination host unreachable.  
Reply from 192.168.43.86: Destination host unreachable.  
Reply from 192.168.43.86: Destination host unreachable.  
Reply from 192.168.43.86: Destination host unreachable.  
Reply from 192.168.43.86: Destination host unreachable.  
Reply from 192.168.43.86: Destination host unreachable.
```

1st Linux

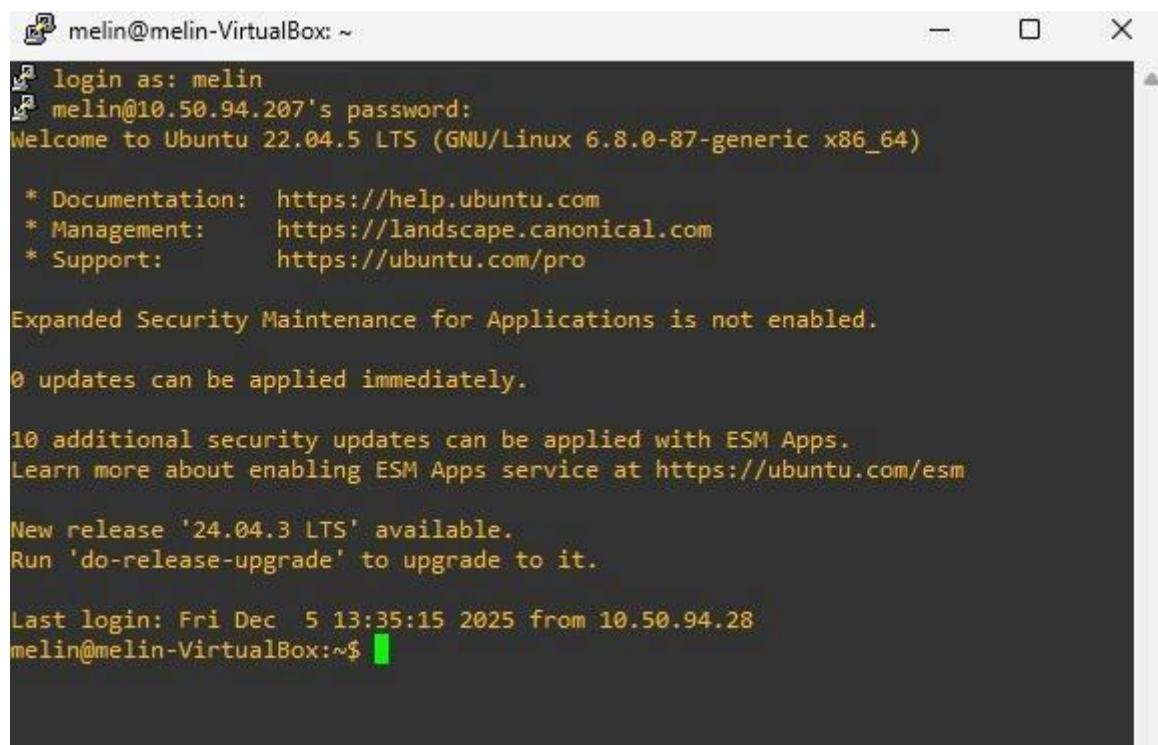
```
iptraf-ng 1.2.1  
- TCP Connections (Source Host:Port) ————— Packets — Bytes Flag Iface —  
  
  
  
  
  
  
  
  
  
TCP:      0 entries ————— Active  
  
ICMP echo req (46 bytes) from 10.50.94.224 to 10.50.94.207 on enp0s8  
ICMP echo req (46 bytes) from 10.50.94.224 to 10.50.94.207 on enp0s8  
ICMP echo req (46 bytes) from 10.50.94.224 to 10.50.94.207 on enp0s8  
ICMP echo req (46 bytes) from 10.50.94.224 to 10.50.94.207 on enp0s8  
ICMP echo req (46 bytes) from 10.50.94.224 to 10.50.94.207 on enp0s8  
Bottom — Time: 0:00 — Drops: 0
```

Friend's linux

```
iptraf-ng 1.2.1  
- TCP Connections (Source Host:Port) ————— Packets — Bytes Flag Iface —  
2404:c0:2460:c1b4:7459:cbe4:69a1:36674 = 7 560 S--- enp0s9  
2620:2d:4002:1::197:80 = 0 0 ---- enp0s9  
  
  
  
  
  
  
TCP:      1 entries ————— Active  
  
ICMP echo rply (28 bytes) from 10.50.94.224 to 10.50.94.207 on enp0s9  
ICMP echo req (46 bytes) from 10.50.94.207 to 10.50.94.224 on enp0s9  
ICMP echo req (46 bytes) from 10.50.94.207 to 10.50.94.224 on enp0s9  
ICMP echo rply (28 bytes) from 10.50.94.224 to 10.50.94.207 on enp0s9  
ICMP echo rply (28 bytes) from 10.50.94.224 to 10.50.94.207 on enp0s9  
Bottom — Time: 0:00 — Drops: 0
```

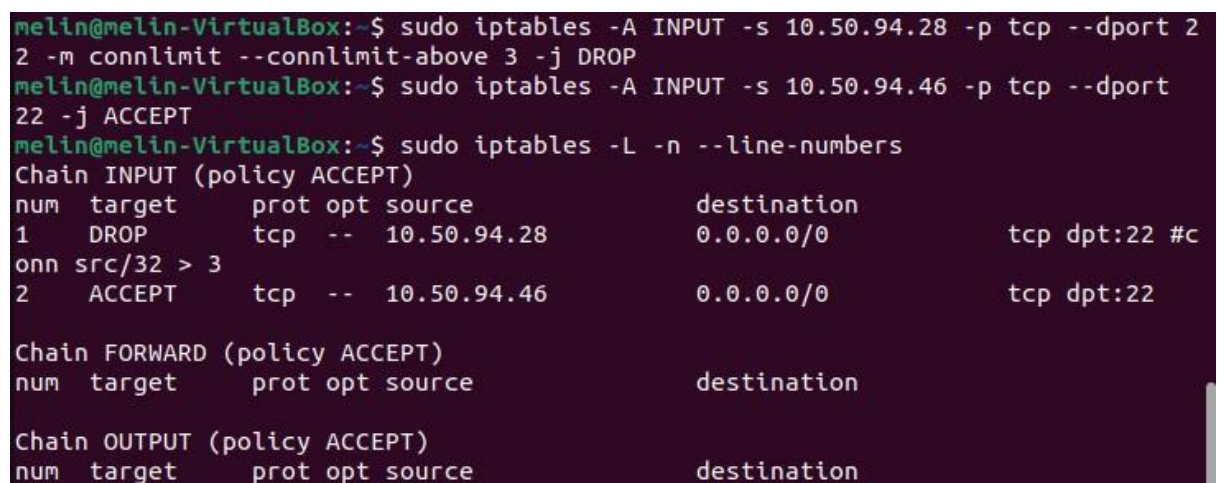

4. Your windows possible to make connection using ssh/putty unlimited connection, but your windows friends (all connection at your windows friend side) only possible to have 3 connections using ssh/putty

1st Windows



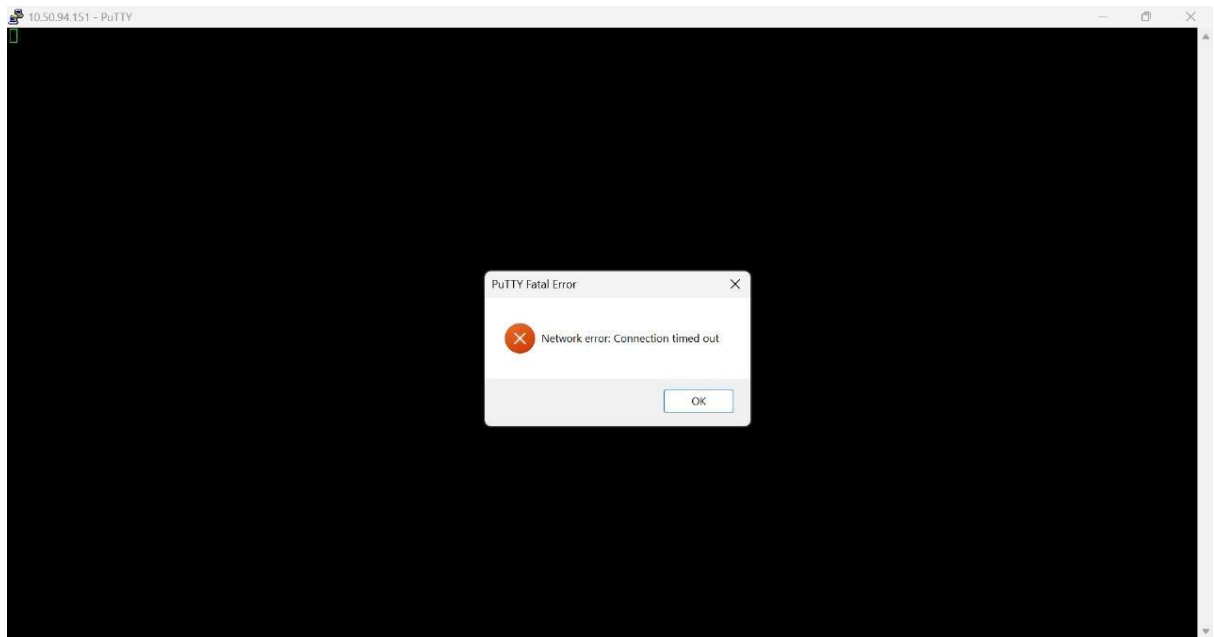
```
melin@melin-VirtualBox: ~  
login as: melin  
melin@10.50.94.207's password:  
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 6.8.0-87-generic x86_64)  
  
* Documentation:  https://help.ubuntu.com  
* Management:    https://landscape.canonical.com  
* Support:       https://ubuntu.com/pro  
  
Expanded Security Maintenance for Applications is not enabled.  
  
0 updates can be applied immediately.  
  
10 additional security updates can be applied with ESM Apps.  
Learn more about enabling ESM Apps service at https://ubuntu.com/esm  
  
New release '24.04.3 LTS' available.  
Run 'do-release-upgrade' to upgrade to it.  
  
Last login: Fri Dec  5 13:35:15 2025 from 10.50.94.28  
melin@melin-VirtualBox:~$
```

The command



```
melin@melin-VirtualBox:~$ sudo iptables -A INPUT -s 10.50.94.28 -p tcp --dport 22 -m connlimit --connlimit-above 3 -j DROP  
melin@melin-VirtualBox:~$ sudo iptables -A INPUT -s 10.50.94.46 -p tcp --dport 22 -j ACCEPT  
melin@melin-VirtualBox:~$ sudo iptables -L -n --line-numbers  
Chain INPUT (policy ACCEPT)  
num  target      prot opt source                destination          tcp dpt:22 #c  
1    DROP        tcp  --  10.50.94.28           0.0.0.0/0            tcp dpt:22  
onn src/32 > 3  
2    ACCEPT     tcp  --  10.50.94.46           0.0.0.0/0            tcp dpt:22  
  
Chain FORWARD (policy ACCEPT)  
num  target      prot opt source                destination  
  
Chain OUTPUT (policy ACCEPT)  
num  target      prot opt source                destination
```

Friend's windows



5. Your linux server only accept ssh connection from your friend windows mac address.
(using command `ipconfig -all` from windows)

Proof that the Linux server only accepts SSH connections from my friend's Windows
MAC address. The SSH login succeeds only from the whitelisted MAC.

```
melin@melin-VirtualBox: ~  
login as: melin  
melin@10.73.13.151's password:  
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 6.8.0-87-generic x86_64)  
  
* Documentation:  https://help.ubuntu.com  
* Management:    https://landscape.canonical.com  
* Support:       https://ubuntu.com/pro  
  
Expanded Security Maintenance for Applications is not enabled.  
  
10 updates can be applied immediately.  
  
10 additional security updates can be applied with ESM Apps.  
Learn more about enabling ESM Apps service at https://ubuntu.com/esm  
  
New release '24.04.3 LTS' available.  
Run 'do-release-upgrade' to upgrade to it.  
  
Last login: Fri Dec  5 13:50:00 2025 from 10.50.94.28  
melin@melin-VirtualBox:~$
```

iptables configuration: Rule added to drop all SSH traffic on port 22, followed by a rule
allowing SSH only from the specific MAC address (14:AC:60:C9:D9:75)

```

melin@melin-VirtualBox:~$ sudo iptables -A INPUT -p tcp --dport 22 -j DROP
melin@melin-VirtualBox:~$ sudo iptables -A INPUT -p tcp --dport 22 -m mac --mac-source 14:AC:60:C9:D9:75 -j ACCEPT
melin@melin-VirtualBox:~$ sudo iptables -L -n --line-numbers
Chain INPUT (policy ACCEPT)
num target      prot opt source                destination              tcp dpt:22
1    DROP          tcp  --  0.0.0.0/0              0.0.0.0/0                tcp dpt:22
2    ACCEPT        tcp  --  0.0.0.0/0              0.0.0.0/0                tcp dpt:22 MA
C14:ac:60:c9:d9:75

Chain FORWARD (policy ACCEPT)
num target      prot opt source                destination

Chain OUTPUT (policy ACCEPT)
num target      prot opt source                destination

```

6. You fulfill all the above firewall rules. Save it. Your lecturer will check this rules

```

nawra@nawra-VirtualBox:~$ sudo iptables -A INPUT -s 10.73.13.55 -p tcp --dport 80 -j DROP
nawra@nawra-VirtualBox:~$ sudo iptables -A INPUT -s 10.73.13.55 -p tcp --dport 22 -j ACCEPT
nawra@nawra-VirtualBox:~$ sudo iptables -A INPUT -s 10.73.13.55 -p tcp --dport 21 -j ACCEPT
nawra@nawra-VirtualBox:~$ sudo iptables -A INPUT -s 10.73.13.231 -p tcp --dport 80 -j ACCEPT
nawra@nawra-VirtualBox:~$ sudo iptables -A INPUT -s 10.73.13.231 -p tcp --dport 22 -j ACCEPT
nawra@nawra-VirtualBox:~$ sudo iptables -A INPUT -s 10.73.13.231 -p tcp --dport 21 -j DROP
nawra@nawra-VirtualBox:~$ sudo iptables -A INPUT -s 10.73.13.17 -p icmp --icmp-type echo-request -j ACCEPT
nawra@nawra-VirtualBox:~$ sudo iptables -A INPUT -p icmp --icmp-type echo-request -j DROP
nawra@nawra-VirtualBox:~$ sudo iptables -L -n --line-numbers
Chain INPUT (policy ACCEPT)
num target      prot opt source                destination              tcp dpt:80
1    DROP          6    --  10.73.13.55            0.0.0.0/0                tcp dpt:22
2    ACCEPT        6    --  10.73.13.55            0.0.0.0/0                tcp dpt:21
3    ACCEPT        6    --  10.73.13.55            0.0.0.0/0                tcp dpt:80
4    ACCEPT        6    --  10.73.13.231           0.0.0.0/0                tcp dpt:22
5    ACCEPT        6    --  10.73.13.231           0.0.0.0/0                tcp dpt:21
6    DROP          6    --  10.73.13.231           0.0.0.0/0                icmp type 8
7    ACCEPT        1    --  10.73.13.17            0.0.0.0/0                icmp type 8
8    DROP          1    --  0.0.0.0/0              0.0.0.0/0

```

Lab 4: Network Address Translation (PPT Week 9)

1. Identify The Valuable Assets And Data / Application. Which Application and data that least important and most important

A. Identifying Valuable Assets and Data

To secure the network, all applications and data are classified into three priority levels:

a. HIGH PRIORITY (Most Critical)

These systems contain highly confidential and sensitive information. Compromise of these systems leads to financial loss, legal consequences, and operational failure.

- Finance Application

- Finance Database
- Medical Records Application
- Medical Records Database

Reason: Contains financial transactions, patient personal data, medical history, and confidential information.

b. MEDIUM PRIORITY

Systems essential for hospital operations but less sensitive than financial and medical record systems.

Laboratory Information System

Paramedical System

Nursing System

Reason: Important for internal workflow, but not as sensitive as financial/medical systems.

c. LOW PRIORITY (Least Critical)

Systems with low confidentiality and accessible by public or general staff.

- Hospital Public Website
- Email Server
- Guest Network Portal

Reason: Contains public information or general communication. Breach impact is low compared to critical systems.

B. Separate between Application Server and Database Server

To prevent lateral movement and minimize damage from attacks, every application must be separated from its database:

Application Server	Database Server	Reason
--------------------	-----------------	--------

Finance App Server	Finance DB Server	Financial data must be isolated
Medical App Server	Medical DB Server	Patient records highly confidential
Laboratory App Server	Lab DB Server	Medium Sensitivity
Public Website Server	Shared DB (Optional)	Lowest sensitivity

Security Principle:

No application and database should run on the same server.

this prevents attackers from gaining direct DB access if the application layer is compromised.

C. Implement firewall segmentation

Application and data that are least important may have one server for many applications (e.g. Hospital web site, possible in the one server with email server)

Application and data that are most important must have dedicated server (e.g. Finance application)

- Every server may have different firewall
- Each server category is assigned to its own firewall zone. A. High Priority Servers Dedicated Firewall

Applies to:

- Finance App & DB
- Medical App & DB

Rules:

- Only Finance Subnet can access Finance Server
- Only Medical Subnet can access Medical Server

- Deny Guest, Nursing, Lab, Paramedical
- Deny all inbound except: HTTP/HTTPS (Web App) & SQL Port (Internal only)
- Implement strict egress filtering (no outbound internet) B. Medium Priority Servers , Semi-Isolated

Applies to:

- Laboratory
- Paramedical
- Nursing

Rules:

- Only respective departmental subnet can access
- Inter-department traffic blocked
- Internet access restricted

Low Priority Servers, Shared Firewall

Applies to:

- Web Server
- Email Server
- Guest Portal

Rules:

- Public internet: only HTTP/HTTPS allowed
- No internal database access
- No privileged traffic allowed

Lab 5: Network Design & Asset Classification (PPT Week 9)

5. Implement Network Segmentation.

- Divide the user into several subnet / subnet mask

- You may divide the subnet based on department (e.g subnet for finance department because no one will access finance application except finance staff. Guest subnet, because guest subnet cannot access many application)
- You may divide the subnet based on floor and building, but you may implement more than one subnet for each floor.
- You may divide the server based on departments (e.g finance application server can be accessed by finance staff.).

1. User Segmentation into Subnets

To prevent over-segmentation and under-segmentation, users are grouped by department.

A. Nursing, Laboratory, Paramedical – 700 Users

- Subnet Mask: 255.255.252.0 (/22) • IP Range: 10.2.12.1 – 10.2.15.254
- Reason: 1022 usable hosts, efficient for 700 users.

B. Guest Network – 1200 Users

- Subnet Mask: 255.255.248.0 (/21)
- IP Range: 192.168.12.1 – 192.168.19.254
- Reason: Large guest population. Must be isolated from all internal networks.

C. Finance + Medical Record – 100 Users

- Subnet Mask: 255.255.255.128 (/25)
- IP Range: 10.2.17.1 – 10.2.17.126
- Reason: Highly secure, limited access, small user base.

2. Server Network Segmentation

Server Group	Subnet	IPs
Finance Servers	10.5.0.0/28	10.5.0.2 & 10.5.0.3
Medical Servers	10.5.0.16/28	10.5.0.18 & 10.5.0.19
Laboratory Servers	10.5.0.32/28	10.5.0.34
DMZ (Web/Email)	172.16.5.0/28	172.16.5.2 & 172.16.5.3

3. Final IP Addressing Plan

A. User Subnets

- Nursing/Lab/Paramedical: 10.2.12.0/22
- Guest: 192.168.12.0/21
- Finance + Medical Record: 10.2.17.0/25

B. Server IP Allocation

Server Name	Role	IP Address
FIN-APP01	Finance Application	10.5.0.2
FIN-DB01	Finance Database	10.5.0.3
MED-APP01	Medical Record App	10.5.0.18
MED-DB01	Medical Database	10.5.0.19
LAB-APP01	Laboratory App	10.5.0.34
WEB01	Public Website	172.16.5.2
MAIL01	Email Server	172.16.5.3

4. Router Configuration Overview

Router functions:

- Inter-VLAN routing
- Connects all subnets
- Routes internal-to-internal
- Passes only allowed traffic to firewalls
- Denies Guest (Internal)
- Denies inter-department traffic

5. Firewall Zones Final Structure

ZONE 1: WAN (Internet) Contains:

- Accessible only to DMZ
- Blocks direct internal access

ZONE 2: DMZ

Contains:

- Public Website
- Email Server
- External-facing services

ZONE 3: INTERNAL SEGMENTED NETWORK

Contains:

- Finance subnet
- Medical subnet
- Nursing subnet
- Lab subnet
- Paramedical subnet
- Server Subnets

Rules:

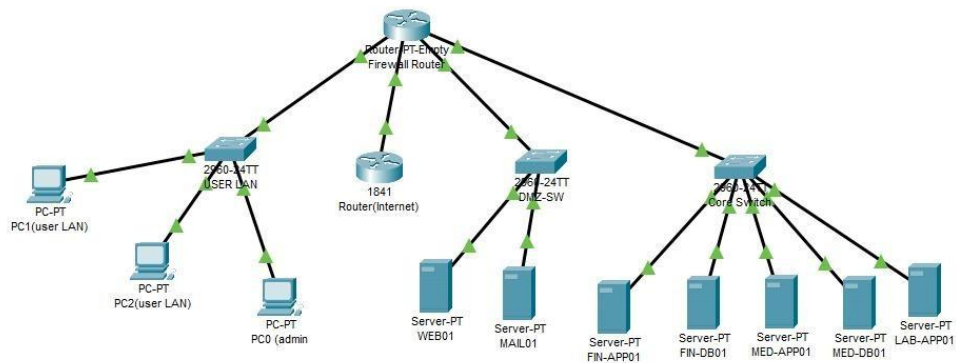
- Finance - Medical = BLOCK
- Guest - ANY Internal = BLOCK
- Nursing - Lab = BLOCK
- Finance - Finance Servers = ALLOW
- Medical - Medical Servers = ALLOW

6. Create the Secure Computer network and fulfill it with :

- IP Address
- Subnet Mask
- Router, Firewall

- Server (give the name for every server)

Topology :



core switch 1

```

Switch>enable
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#vlan 10
Switch(config-vlan)# name FINANCE
Switch(config-vlan)#vlan 20
Switch(config-vlan)# name MEDICAL
Switch(config-vlan)#vlan 30
Switch(config-vlan)# name LAB
Switch(config-vlan)#exit
Switch(config)#interface fa0/1
Switch(config-if)# switchport mode access
Switch(config-if)# switchport access vlan 10
Switch(config-if)#
Switch(config-if)#interface fa0/2
Switch(config-if)# switchport mode access
Switch(config-if)# switchport access vlan 10
Switch(config-if)#
Switch(config-if)#interface fa0/3
Switch(config-if)# switchport mode access
Switch(config-if)# switchport access vlan 20
Switch(config-if)#
Switch(config-if)#interface fa0/4
Switch(config-if)# switchport mode access
Switch(config-if)# switchport access vlan 20
Switch(config-if)#
Switch(config-if)#interface fa0/5
Switch(config-if)# switchport mode access
Switch(config-if)# switchport access vlan 30
Switch(config-if)#exit
Switch(config)#interface fa0/24
Switch(config-if)# switchport mode trunk
  
```

Internet router

```
Router>enable
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#ip route 10.0.0.0 255.0.0.0 200.200.200.1
```

Main router (command 1)

```
Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

Router(config-if)#interface g0/1
%Invalid interface type and number
Router(config)# ip address 200.200.200.1 255.255.255.0
^
% Invalid input detected at '^' marker.

Router(config)#interface g1/0
Router(config-if)# ip address 200.200.200.1 255.255.255.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet1/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet1/0, changed state to up

Router(config-if)#int g2/0
Router(config-if)#ip address 172.16.5.1 255.255.255.240
Router(config-if)# no shut

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet2/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet2/0, changed state to up

Router(config-if)#int g3/0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet3/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet3/0, changed state to up

Router(config-if)#int g3/0.10
Router(config-subif)#
%LINK-5-CHANGED: Interface GigabitEthernet3/0.10, changed state to up
```

Main router (command 2)

```
Router(config-subif)#encapsulation dot1q 10
Router(config-subif)# ip address 10.5.0.1 255.255.255.240
Router(config-subif)#int g3/0.20
Router(config-subif)#
%LINK-5-CHANGED: Interface GigabitEthernet3/0.20, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet3/0.20, changed state to up

Router(config-subif)#encapsulation dot1q 20
Router(config-subif)# ip address 10.5.0.17 255.255.255.240
Router(config-subif)#int g3/0.30
Router(config-subif)#
%LINK-5-CHANGED: Interface GigabitEthernet3/0.30, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet3/0.30, changed state to up

Router(config-subif)#encapsulation dot1q 30
Router(config-subif)# ip address 10.5.0.33 255.255.255.240
Router(config-subif)#no shutdown
Router(config-subif)#exit
Router(config)#ip route 0.0.0.0 0.0.0.0 200.200.200.2
```

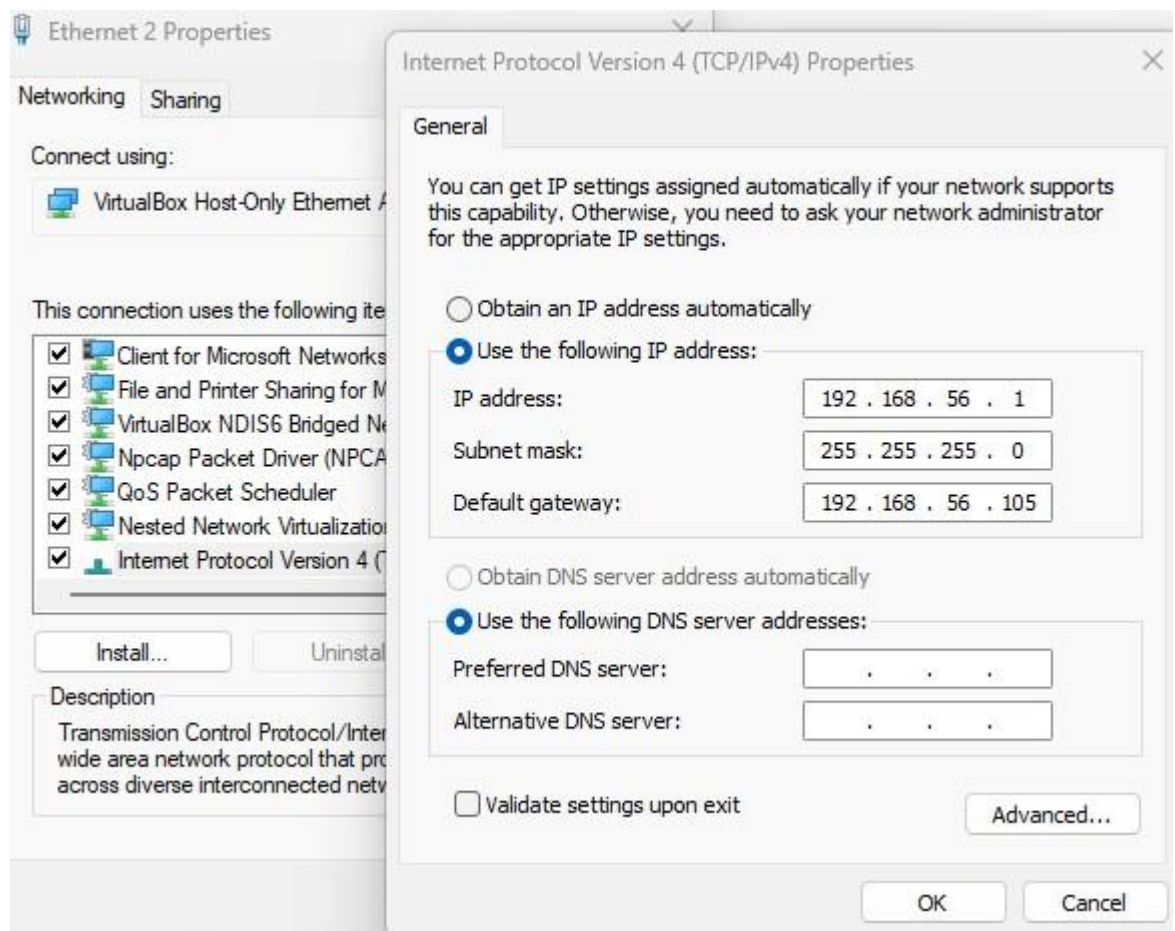
Main router (command 3)

```
Router>enable
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#access-list 1 permit 192.168.10.0 0.0.0.255
Router(config)#access-list 1 permit 10.5.0.0 0.0.0.255
Router(config)#access-list 1 permit 172.16.5.0 0.0.0.255
Router(config)#ip nat inside source list 1 interface g1/0 overload
Router(config)#interface g1/0
Router(config-if)#ip nat outside
Router(config-if)#interface g0/0
Router(config-if)# ip nat inside
Router(config-if)#interface g2/0
Router(config-if)# ip nat inside
Router(config-if)#interface g3/0
Router(config-if)# ip nat inside
Router(config-if)#access-list 110 permit ip 192.168.10.0 0.0.0.255 10.5.0.0 0.0.0.15
Router(config)#access-list 110 deny ip any 10.5.0.0 0.0.0.15
Router(config)#access-list 120 permit ip host 192.168.10.50 10.5.0.16 0.0.0.15
Router(config)#access-list 120 deny ip any 10.5.0.16 0.0.0.15
Router(config)#access-list 130 permit ip host 192.168.10.99 host 10.5.0.34
Router(config)#access-list 130 deny ip any host 10.5.0.34
Router(config)#access-list 140 permit tcp any host 172.16.5.2 eq 80
Router(config)#access-list 140 permit tcp any host 172.16.5.3 eq 25
Router(config)#interface g3/0
Router(config-if)#ip access-group 110 in
Router(config-if)# ip access-group 120 in
Router(config-if)# ip access-group 130 in
Router(config-if)#interface g2/0
Router(config-if)#ip access-group 140 in
```

Lab 6: Secure Network Implementation (PPT Week 10)

1. Computer from internet (your friend) possible to access web server using firewall ip address

1. Windows virtualbox adapter configuration



Linux configuration

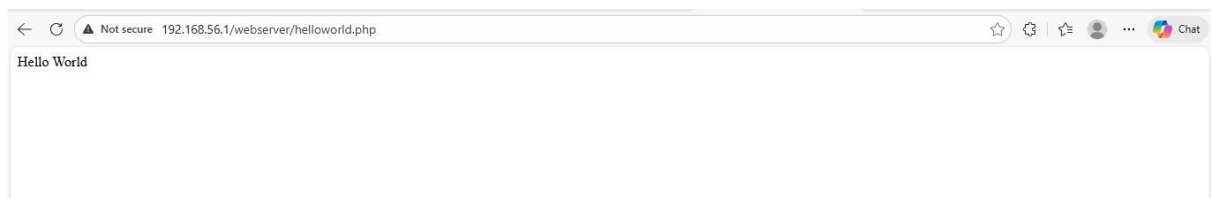

```

enp0s8: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
        inet 192.168.56.105 netmask 255.255.255.0 broadcast 192.168.56.255
        inet6 fe80::338:2d6d:5453:21b prefixlen 64 scopeid 0x20<link>
        ether 08:00:27:aa:ae:48 txqueuelen 1000 (Ethernet)
        RX packets 2343 bytes 354146 (354.1 KB)
        RX errors 0 dropped 0 overruns 0 frame 0
        TX packets 363 bytes 43109 (43.1 KB)
        TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

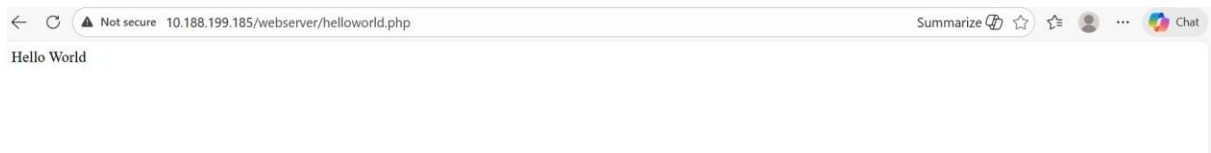
enp0s9: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
        inet 10.188.199.185 netmask 255.255.255.0 broadcast 10.188.199.255
        inet6 fe80::67a8:e55a:bc52:e15b prefixlen 64 scopeid 0x20<link>
        ether 08:00:27:e1:b9:f0 txqueuelen 1000 (Ethernet)
        RX packets 5011 bytes 841726 (841.7 KB)
        RX errors 0 dropped 0 overruns 0 frame 0
        TX packets 1898 bytes 222232 (222.2 KB)
        TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

```

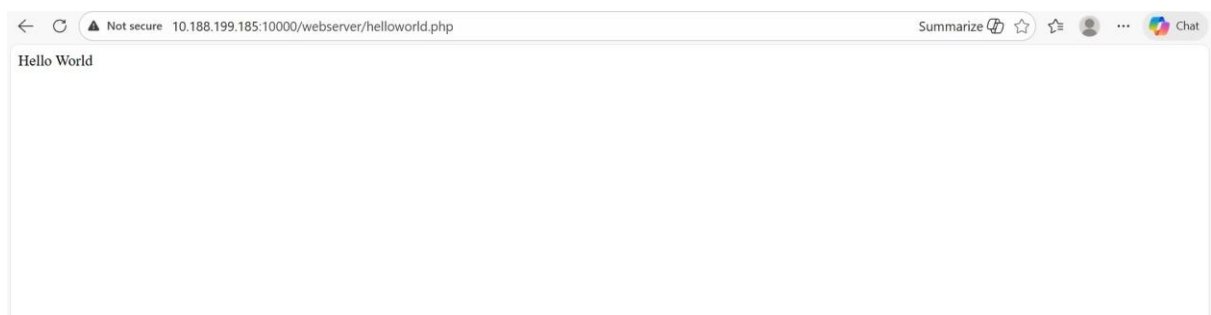
3. My windows webserver



3. Friends access webserver through linux firewall ip address



4. Friends access webserver through linux firewall ip address using port 10000.png



Linux command (prerouting 1)

```

melin@melin-VirtualBox:~$ nano /proc/sys/net/ipv4/ip_forward
melin@melin-VirtualBox:~$ sudo nano /proc/sys/net/ipv4/ip_forward
[sudo] password for melin:
melin@melin-VirtualBox:~$ sudo nano /proc/sys/net/ipv4/ip_forward
melin@melin-VirtualBox:~$ echo 1 > /proc/sys/net/ipv4/ip_forward
bash: /proc/sys/net/ipv4/ip_forward: Permission denied
melin@melin-VirtualBox:~$ sudo echo 1 > /proc/sys/net/ipv4/ip_forward
bash: /proc/sys/net/ipv4/ip_forward: Permission denied
melin@melin-VirtualBox:~$ echo 1 | sudo tee /proc/sys/net/ipv4/ip_forward
1
melin@melin-VirtualBox:~$ cat /proc/sys/net/ipv4/ip_forward
melin@melin-VirtualBox:~$ sudo iptables -t nat -A PREROUTING -p tcp -d 10.188.19
9.185 --dport 80 -j DNAT --to 192.168.56.1:80
melin@melin-VirtualBox:~$ sudo iptables -t nat -L
Chain PREROUTING (policy ACCEPT)
target     prot opt source                destination
DNAT       tcp  --  anywhere               melin-VirtualBox      tcp dpt:http to:19
2.168.56.1:80

Chain INPUT (policy ACCEPT)
target     prot opt source                destination

Chain OUTPUT (policy ACCEPT)

```

Linux command (prerouting 2)

```

melin@melin-VirtualBox:~$ sudo iptables -t nat -A PREROUTING -p tcp -d 10.188.19
9.185 --dport 10000 -j DNAT --to 192.168.56.1:80
melin@melin-VirtualBox:~$ sudo iptables -t nat -L
Chain PREROUTING (policy ACCEPT)
target     prot opt source                destination
DNAT       tcp  --  anywhere               melin-VirtualBox      tcp dpt:webmin to:
192.168.56.1:80

Chain INPUT (policy ACCEPT)
target     prot opt source                destination

Chain OUTPUT (policy ACCEPT)
target     prot opt source                destination

Chain POSTROUTING (policy ACCEPT)
target     prot opt source                destination
melin@melin-VirtualBox:~$ sudo iptables -t nat -F

```

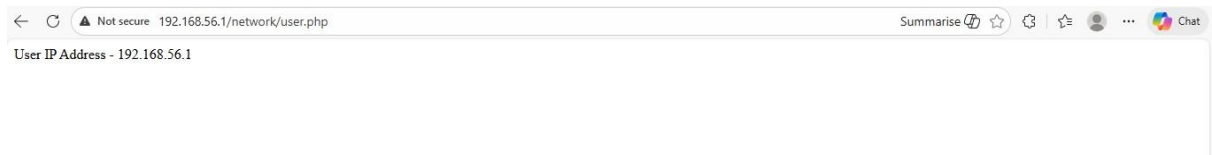
2. Users possible to access to internet (your friend computer) through firewall. Your friend computer will see the firewall ip (firewall hides your computer ip) Linux command

```

melin@melin-VirtualBox:~$ sudo iptables -t nat -A POSTROUTING -o enp0s9 -j SNAT
--to 10.188.199.185

```

Friends windows webserver



I access my friend webserver through my friends address

